

Web Technology

By

Phul Babu Jha

Course Title: Web Technology

Course No: CSC318

Nature of the Course: Theory + Lab

Year: Third, Semester: Fifth

Full Marks: 60+20+20

Pass Marks: 24+8+8

Credit Hours: 3

Course Description: This course covers the fundamental concepts of HTML, CSS, JavaScript, XML, and PHP.

Course Objectives: The main objective of this course is to provide basic knowledge of web design using HTML and CSS, client side scripting using JavaScript, handling web data using XML and server side scripting using PHP.

Detail Syllabus

Chapters / Units	Teaching Methodology	Teaching Hours
Unit I: Introduction 1.1.Web Basics: Internet, Intranet, WWW, Static and Dynamic Web Page; Web Clients; Web Servers 1.2.Client Server Architecture: Single Tier, Two-Tier, Multi-Tier 1.3.HTTP: HTTP Request and Response; URL 1.4.Client Side Scripting, Server Side Scripting, 1.5.Web 1.0, Web 2.0	Class Lecture	3 Hours

Unit II: Hyper Text Markup Language

- 2.1. Introduction to HTML; Elements of HTML Document; HTML Elements and HTML Attributes, Headings, Paragraph, Division , Comments in HTML
- 2.2. Formatting: b, i, small, sup, sub; Spacing: Pre, Br; Formatting Text Phrases: span, strong, tt
- 2.3. Image element; Anchors; Lists: Ordered and Unordered and Definition;
- 2.4. Tables(table, tr, td colspan, rowspan, thead, tbody,tfoot); Frames (iframe)
- 2.5. Forms: Form Elements, ID attributes, Class Attributes of HTML Elements

**Class Lecture
+
Lab Session**

10 Hours

- 2.6. Meta Tag, Audio (src, controls, autoplay, preload, loop), Video (src, controls, autoplay, poster, loop), Canvas, Main, Section, Article, Header, Footer, Aside, Nav, Figure Tags
- 2.7. HTML Events: Window Events, Form Element Events, Keyboard Events, Mouse Events

Unit V: AJAX and XML	Class Lecture + Lab Session	7 Hours
5.1. Basics of AJAX		
5.2. Introduction to XML and its Application		
5.3. Syntax Rules for creating XML document		
5.4. XML Elements; XML Attributes; XML Tree		
5.5. XML Namespace		
5.6. XML schema languages: Document Type Definition(DTD), XML Schema Definition (XSD); XSD Simple Types, XSD Attributes; XSD Complex Types		
5.7. XML Style Sheets (XSLT)		
5.8. Xquery		

Unit VI: Server Side Scripting using PHP	Class Lecture + Lab Session	8 Hours
6.1. PHP Syntax		
6.2. Variables, Data Types , Strings, Constants		
6.3. Operators, Control structure, Functions, Array		
6.4. Creating Class and Objects		
6.5. PHP Forms, Accessing Form Elements, Form Validation, Events		
6.6. Cookies and Sessions		
6.7. Working with PHP and MySQL, Connecting to Database, Creating, Selecting, Deleting, Updating Records in a table, Inserting Multiple Data		
6.8. Introduction to CodeIgniter, Laravel, Wordpress (Basic concepts and Installation)		

Laboratory Work Manual

The laboratory work includes creating web pages and applications with using HTML, CSS, JavaScript, XML, and PHP. Student should write programs and prepare lab sheet for all of the units in the syllabus. The instructors have to prepare lab sheets for individual units covering the concept of the units as per the requirement. All of the lab reports should be evaluated during the corresponding weeks of hands on practice. The lab session for above chapters should be as per following description however the depth of lab works are not limited to the below mentioned contents only. Based on the lab works in each of the units, students have to prepare a web based application, using above mentioned technologies, as a project work.

Unit II: Unit 2: Hyper Text Markup Language (10 Hrs.)

- Write HTML scripts for illustrating all of the concepts mentioned in this unit.

Unit III: Cascading Style Sheets (8 Hrs.)

- Write CSS scripts for implementing the concepts mentioned in this unit like;
 - o Inline, internal and external css for Backgrounds; Borders; Text; Font; List; Table, HTML elements etc.
 - o Class, id, box models, Display layouts
 - o CSS3 concepts like box shadow, text effects etc.
 - o Responsive Designs using media queries (media types, viewports)
 - o Using Bootstrap

Unit IV: Client Side Scripting with JavaScript (9 Hrs.)

- Write programs/scripts for illustrating the concepts of JavaScript mentioned in this unit as
 - o Operators, Flow Controls, Looping, Functions; Popup Boxes
 - o Arrays
 - o Built-in Objects: Window, String, Number, Boolean, Date, Math, RegExp, Form
 - o Event Handling and Form Validation, Error Handling,
 - o Handling Cookies,
- Write programs/scripts for implementing
 - o Basics of jQuery Syntax;
 - o jQuery Selectors, Events and Effects;
 - o Basics of JSON

Unit V: AJAX and XML (7 Hrs.)

- Write programs/scripts for implementing
 - o Basics of AJAX; Using XMLHttpRequest Object
 - o XML Files with XML Elements and XML Attributes
 - o XML schema languages: Document Type Definition(DTD), XML Schema Definition (XSD)
 - o XML Style Sheets (XSLT)

❑ Tour of Web Technology:-

Where we are and What we are going to learn ?

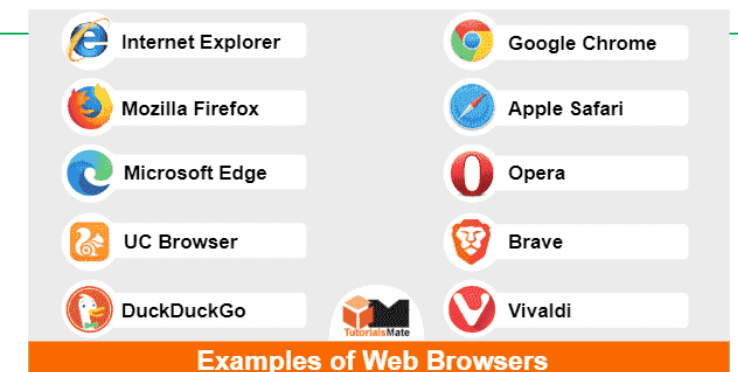
- Web Technology refers to the various tools and techniques that are utilized in the process of communication between different types of devices over the internet. A web browser is used to access web pages.
- Web browsers can be defined as programs that display text, data, pictures, animation, and video on the Internet. Hyperlinked resources on the World Wide Web can be accessed using software interfaces provided by Web browsers.

❑ Web Technology can be classified into the following sections:

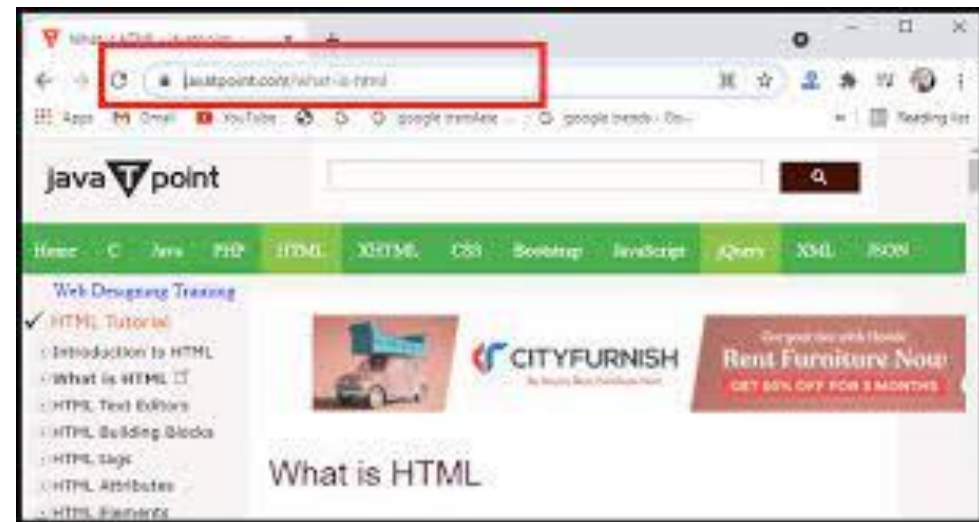
- **World Wide Web (WWW):** The World Wide Web is based on several different technologies : Web browsers, Hypertext Markup Language (HTML) and Hypertext Transfer Protocol (HTTP).
- **Web Browser:** The web browser is an application software to explore www (World Wide Web). It provides an interface between the server and the client and requests to the server for web documents and services.



Compiled By :- P. B. Jha (B.SC.CSIT First –Web Technology)



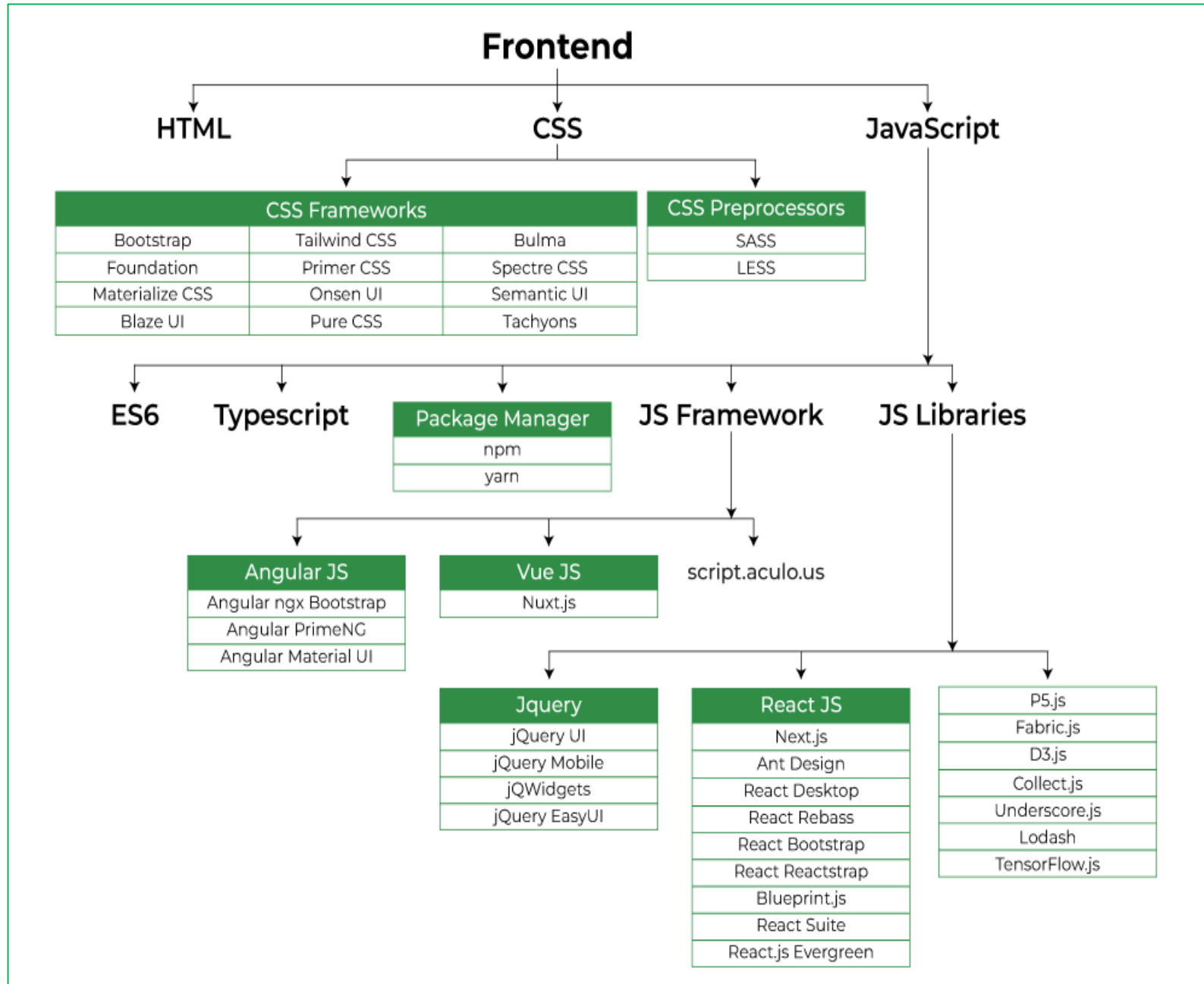
- **Web Server:** Web server is a program which processes the network requests of the users and serves them with files that create web pages. This exchange takes place using Hypertext Transfer Protocol (HTTP).
- **Web Pages:** A webpage is a digital document that is linked to the World Wide Web and viewable by anyone connected to the internet has a web browser.
- **Web Development:** Web development refers to the building, creating, and maintaining of websites. It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet i.e. websites.



❑ Web Development can be classified into two ways:

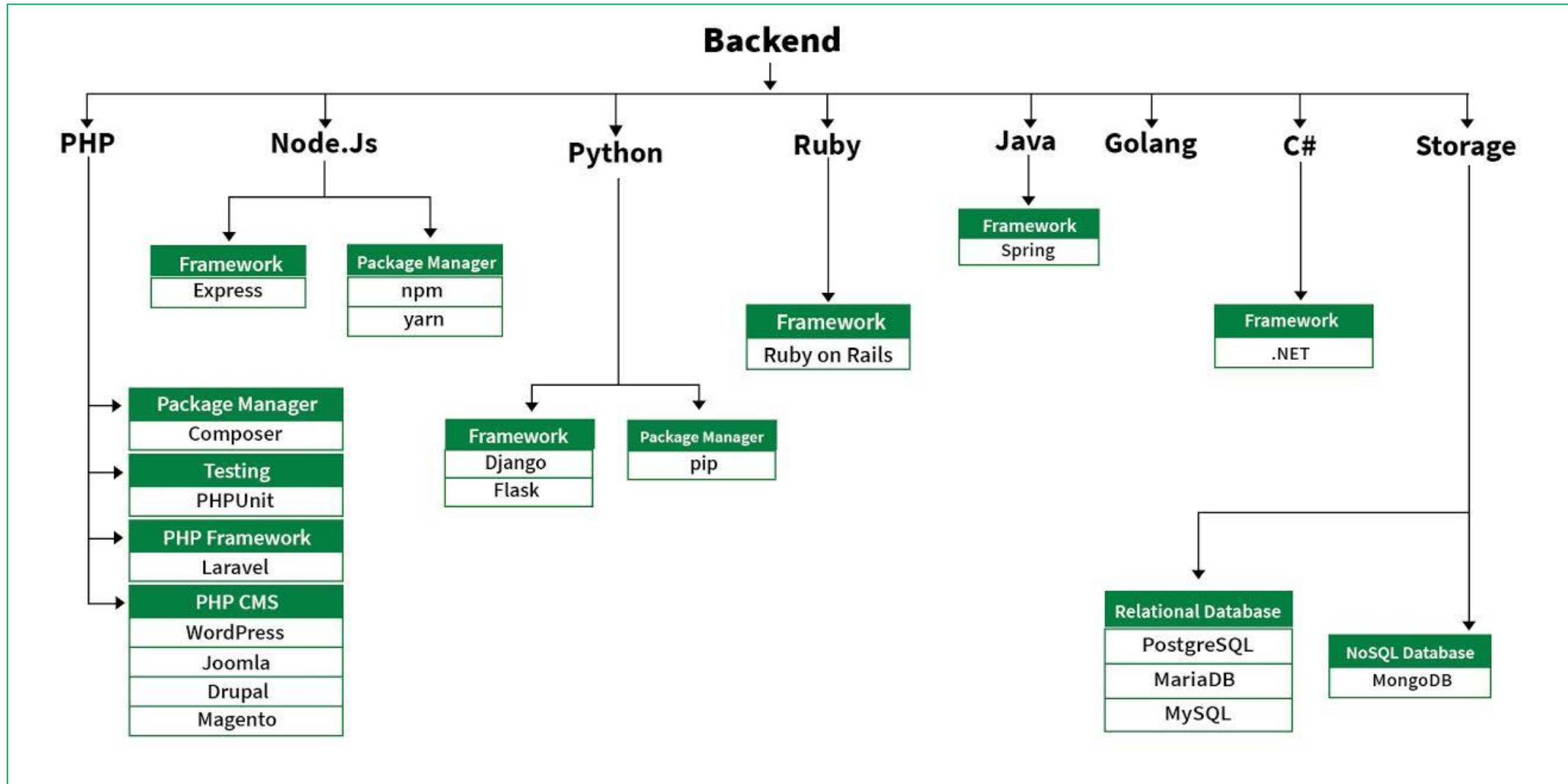
1. Frontend Development:

The part of a website that the user interacts directly is termed as front end. It is also referred to as the 'client side' of the application.



There are many other languages through which one can do front-end development depending upon the framework for example Flutter user Dart, React uses JavaScript and Django uses Python, and much more.

2.Backend Development: Backend is the server side of a website. It is the part of the website that users cannot see and interact. It is the portion of software that does not come in direct contact with the users. It is used to store and arrange data.



- ❑ **Data Format:** Format of data is used by web applications to communicate with each other. It is light weight text based data interchange format which means, it is simpler to read and write.
- Below are two common data formats used in web development.
 1. **XML:** Extensible Markup Language (XML) is a markup language that defines a set of rules for encoding documents in a format that is both human-readable and machine-readable.
 2. **JSON:** JSON or JavaScript Object Notation is a format for structuring data.
- ❑ **API:** API is an abbreviation for Application Programming Interface which is a collection of communication protocols and subroutines used by various programs to communicate between them.
- ❑ **Web Protocols:** Web protocols are set of rules followed by everyone communicating over the web.
- ❑ **HTTP:** The Hypertext Transfer Protocol (HTTP) is designed to enable communications between clients and servers. HTTP works as a request-response protocol between a client and server.
A web browser may be the client, and an application on a computer that hosts a web site may be the server.

❑ Other Protocols:

- TCP/IP Model
- UDP
- FTP
- SMTP
- SOAP

❑ **Graphics:** Graphical elements are one of the key feature of any webpage. They can be used to convey important points better than text does and beautify the webpage.

❑ **Canvas:** The HTML “canvas” element is used to draw graphics via JavaScript.

SVG: SVG stands for Scalable Vector Graphics.

It basically defines vector-based graphics in XML format.

❑ Basic Requirement to Build A web site [Front End]



VectorStock

VectorStock.com/26574321

- Must needed to build web site
- Provide body design



- To beautify the web site
- To make stylish body design



How to design a car with rough rendering



- Brain
- To assign how to work



How to design a car with rough rendering

Engine

Unit I: Introduction

1.1.Web Basics: Internet, Intranet, WWW, Static and Dynamic Web Page; Web Clients; Web Servers

1.2.Client Server Architecture: Single Tier, Two-Tier, Multi-Tier

1.3.HTTP: HTTP Request and Response; URL

1.4.Client Side Scripting, Server Side Scripting,

1.5.Web 1.0, Web 2.

1.1.Web Basics: Internet, Intranet, WWW, Static and Dynamic Web Page; Web Clients; Web Servers

❑ Introduction to Internet:-

- A global computer network providing a variety of information and communication facilities, consisting of interconnected networks using standardized communication protocols. "the guide is also available on the Internet"
- The Internet is the global system of interconnected computer networks that use the Internetprotocol suite (TCP/IP) to link devices worldwide.
- It is a network of networks that consists of private, public, academic, business, and government networks of local to global scope, linked by a broad array of electronic, wireless, and optical networking technologies.
- The Internet carries a vast range of information resources and services.

History of Internet

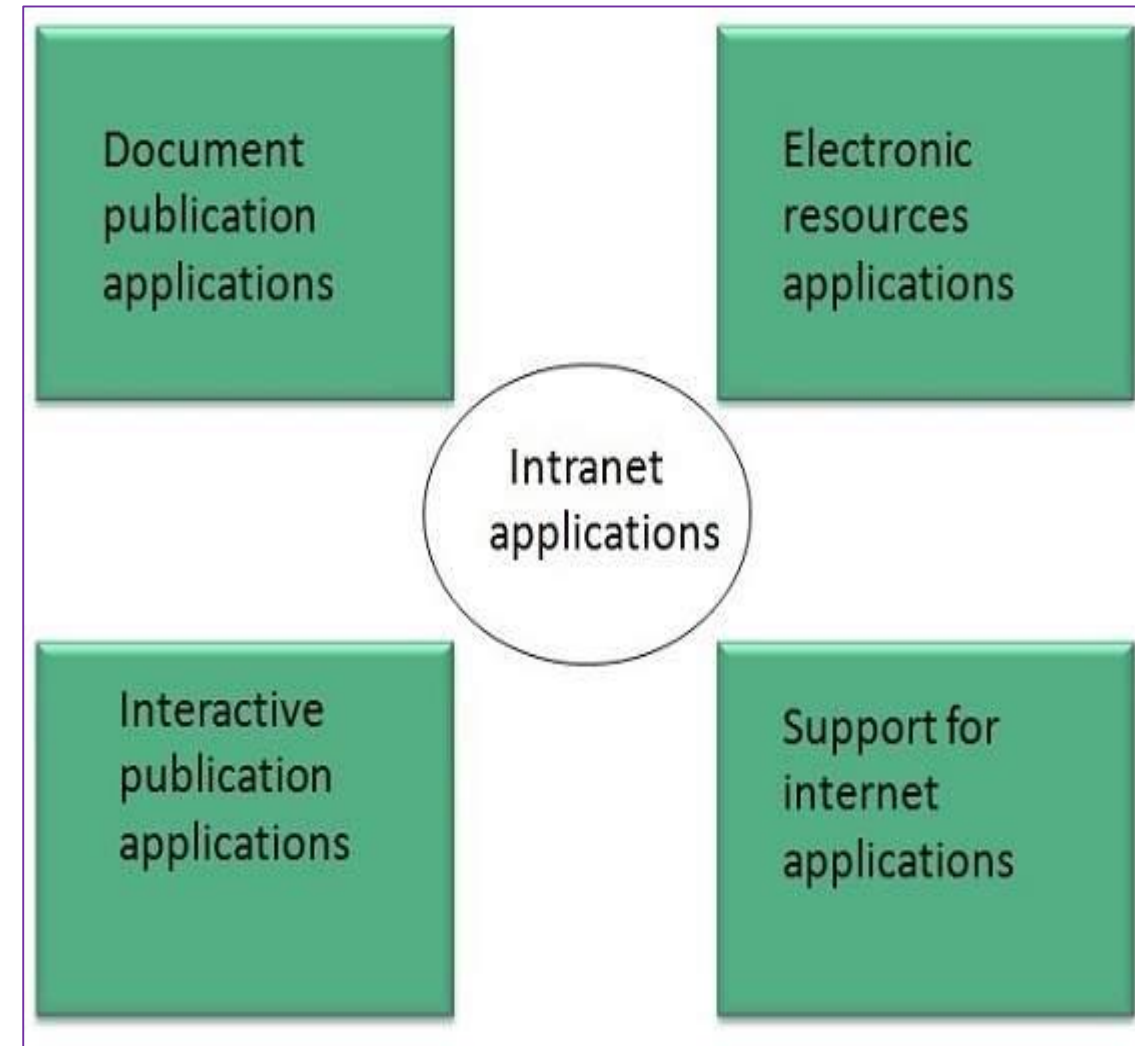
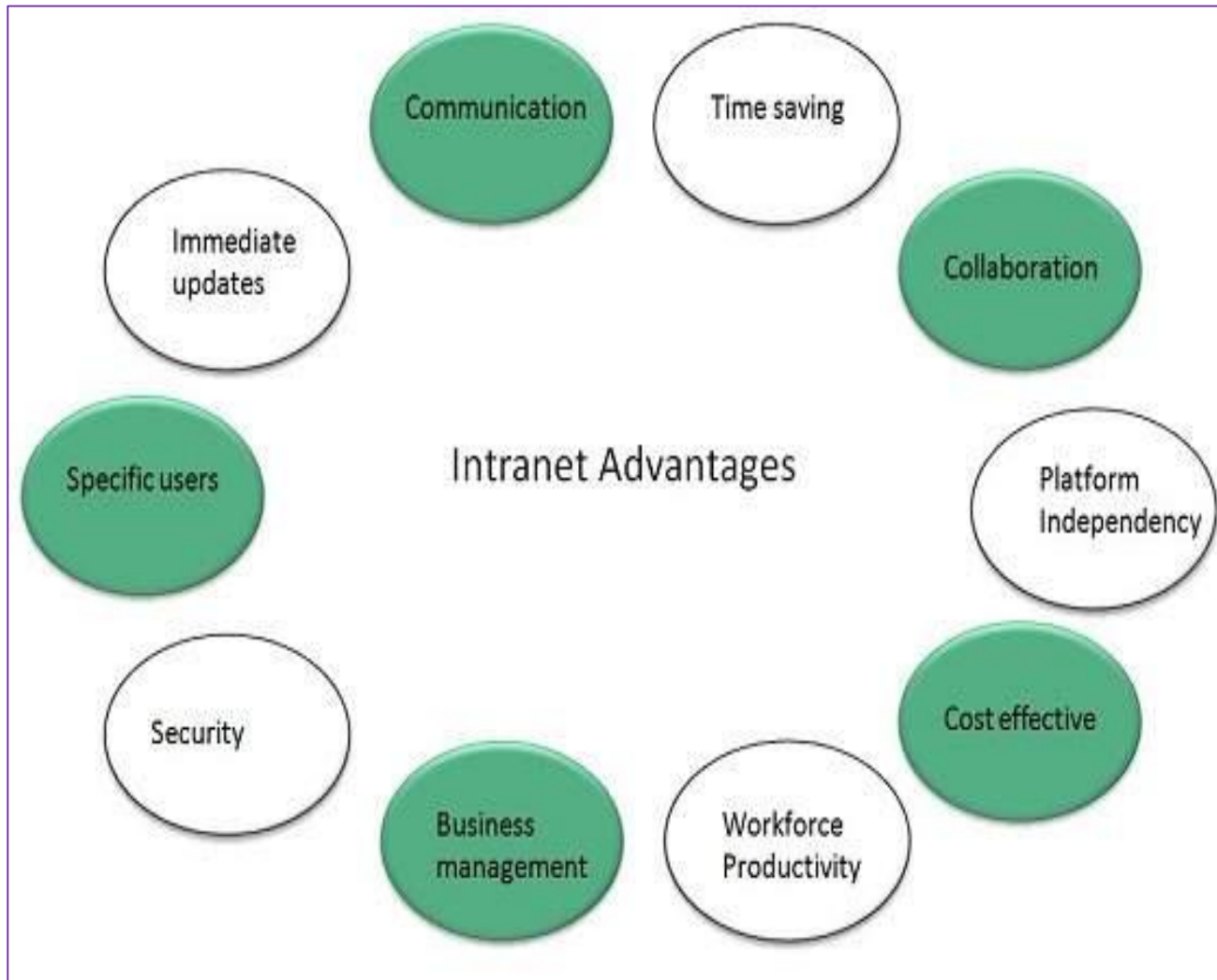
- This marvelous tool has quite a history that holds its roots in the cold war scenario. A need was realized to connect the top universities of the United States so that they can share all the research data without having too much of a time lag. This attempt was a result of Advanced Research Projects Agency (ARPA) which was formed at the end of 1950s just after the Russians had climbed the space era with the launch of Sputnik. After the ARPA got success in 1969, it didn't take the experts long to understand that how much potential can this interconnection tool have. In 1971 Ray Tomlinson made a system to send electronic mail. This was a big step in the making as this opened gateways for remote computer accessing i.e.telnet.
- During all this time, rigorous paper work was being done in all the elite research institutions. From giving every computer an address to setting out the rules, everything was getting penned down. 1973 saw the preparations for the vital TCP/IP and Ethernet services. At the end of 1970s, Usenet groups had surfaced up. By the time the 80s had started, IBM came up with its PC based on Intel 8088 processor which was widely used by students and universities for it solved the purpose of easy computing. By 1982, the Defense Agencies made the TCP/IP compulsory and the term —internet|| was coined. The domain name services arrived in the year 1984 which is also the time around which various internet based marked their debut. A worm, or a rust the computers, attacked in 1988 and disabled over 10% of the computer systems all over the world. While most of the researchers regarded it as an opportunity to enhance computing as it was still in its juvenile phase, quite a number of computer companies became interested in dissecting the cores of the malware which resulted to the formation Computer Emergency Rescue Team (CERT). Soon after the world got over with the computer worm, World Wide Web came into existence. Discovered by Tim Berners-Lee, World Wide Web was seen as a service to connect documents in websites using hyperlinks

Intranet

- Intranet is defined as private network of computers within an organization with its own server and firewall.

Moreover we can define Intranet as:

- Intranet is system in which multiple PCs are networked to be connected to each other. PCs in intranet are not available to the world outside of the intranet.
- Usually each company or organization has their own Intranet network and members/employees of that company can access the computers in their intranet.
- Every computer in internet is identified by a unique IP address.
- Each computer in Intranet is also identified by a IP Address, which is unique among the computers in that Intranet.



World Wide Web

- The World Wide Web (abbreviated WWW or the Web) is an information space where documents and other web resources are identified by Uniform Resource Locators (URLs), interlinked by hypertext links, and can be accessed via the Internet.
- English scientist **Tim Berners-Lee** invented the World Wide Web in 1989. He wrote the first web browser computer program in 1990 while employed at CERN in Switzerland.
- The Web browser was released outside CERN in 1991, first to other research institutions starting in January 1991 and to the general public on the Internet in August 1991.
- The World Wide Web has been central to the development of the Information Age and is the primary tool billions of people use to interact on the Internet.
- Web pages are primarily text documents formatted and annotated with Hypertext Markup Language (HTML). In addition to formatted text, web pages may contain images, video, audio, and software components that are rendered in the user's web browser as coherent pages of multimedia content.

- Embedded hyperlinks permit users to navigate between web pages. Multiple web pages with a common theme, a common domain name, or both, make up a website. Website content can largely be provided by the publisher, or interactively where users contribute content or the content depends upon the users or their actions. Websites may be mostly informative, primarily for entertainment, or largely for commercial, governmental, or non-governmental organizational purposes
- WWW is another example of client/server computing. Each time a link is followed, the client is requesting a document (or graphic or sound file) from a server (also called a Web server) that's part of the World Wide Web that "serves" up the document. The server uses a protocol called HTTP or Hyper Text Transport Protocol. The standard for creating hypertext documents for the WWW is Hyper Text Markup Language or HTML. HTML essentially codes plain text documents so they can be viewed on the Web

Static and Dynamic Web Page

❑ What is a Dynamic Webpage?

- Dynamic websites are those that generate webpages in real-time. Web scripting code, such as PHP or ASP, is used on these pages. The Web server parses the code, and the generated HTML is transmitted to the client's browser when a client requests a dynamic page.
- As dynamic websites are easier to maintain than static websites, the majority of large websites are dynamic. Static pages have their own set of content, hence they must be manually opened, changed, and published whenever they are updated.
- Dynamic pages, on the other hand, use a database to store the data. As a result, the webmaster may need to edit a database record to change the content of a dynamic page. This is particularly useful for sites with hundreds or thousands of pages. It also allows numerous users to update a website's content without having to change the page layout.
- Database-driven websites are dynamic websites that access information from a database.

❑ What is a Static Webpage?

- Static webpages are relatively straightforward. They are written in HTML, JavaScript, CSS, and other languages. When a server receives a request from a static webpage, it transmits the response to the client without performing any more processing.
- In some ways, a static webpage is just a simple information provider. Designers frequently combine text and images controlled by HTML elements to create something that resembles a newspaper page. It has typography and layout, but it does not change with each load.
- Dynamic webpages, in contrast, use deep-coded controls and forms, allowing the pages to react differently depending on the user or the situation. A dynamic webpage, for example, can consult a database to learn more about a user's identity and history or to display personalized elements such as the user's name or saved preferences. A static Web page, on the other hand, does not allow for this level of personalization.
- A static webpage is created with HTML code and has the same look and feel for all users, independent of their identification or other considerations. Dynamic webpages, which may have configurable information based on a user's identification or other criteria, are more difficult to code and assemble.

Difference between Static and Dynamic Webpages

Key	Static Webpage	Dynamic Webpage
Definition	Static webpages are generally simple HTML written pages in which all the data is static in nature and it does not get changed until someone changes it manually.	Dynamic webpages are the pages written in some more complex language such as ASP.NET in which the data is rendered after some interpretation. It can produce distinctive content for different calls.
Complexity	Static websites are simple because all the data is static and do not require any interpretation before rendering.	Dynamic webpages, on other hand, interpret the data which makes the process dynamic in nature. Hence, dynamic webpages are complex as compared to static webpages.
Language used	Static webpages are generally written in simpler languages such as HTML, JavaScript, CSS, etc.	Dynamic webpages are written in more complex languages such as CGI, AJAX, ASP, ASP.NET, etc.
Rendered Data	For static webpages, data does not change until someone changes it manually. Data is static in nature.	The data on a Dynamic webpage first interoperate at the server side, so it does not remain same on every call and this makes the data dynamic in nature.
Time	It is obvious that Static webpages load faster because the data is static.	Dynamic webpages take comparatively more time to load as compared to static webpages.
Database	In Static webpages, generally there is no involvement of a database for data redecoration.	Dynamic webpages use a database for data redecoration.

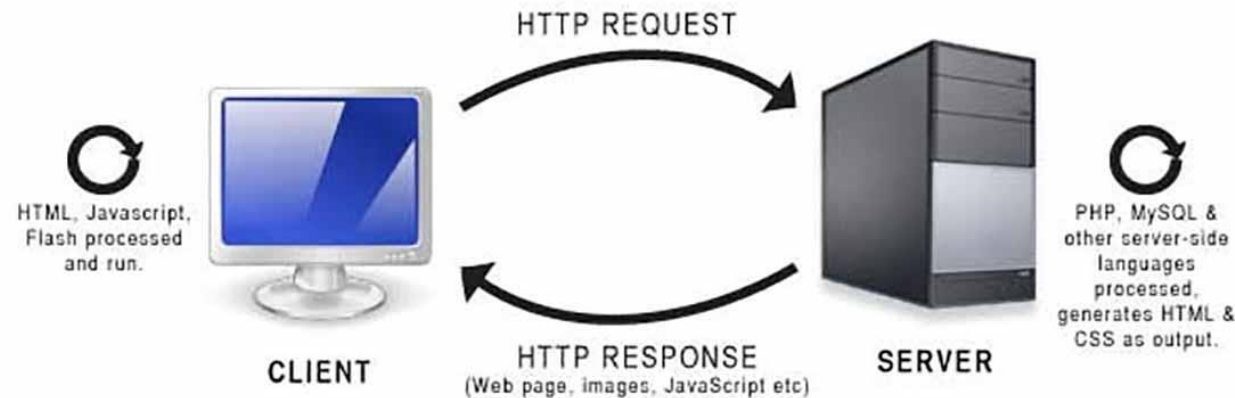
Conclusion

- To conclude, the data on a static website is fixed and it displays the same content to every user, because the web server in this case does not alter the data before sending it as a response to the user. Static website are easy to build and simple to maintain. Most of the read-only websites like brochure websites, portfolio websites, etc., are static in nature.
- A dynamic website, on the other hand, is one where the content is pulled on-the-go, which is why it may show different content to different users. Dynamic websites are highly flexible and scalable. They can offer a better engaging experience by modifying the content as per the requirements of different users.

Web Clients; Web Servers

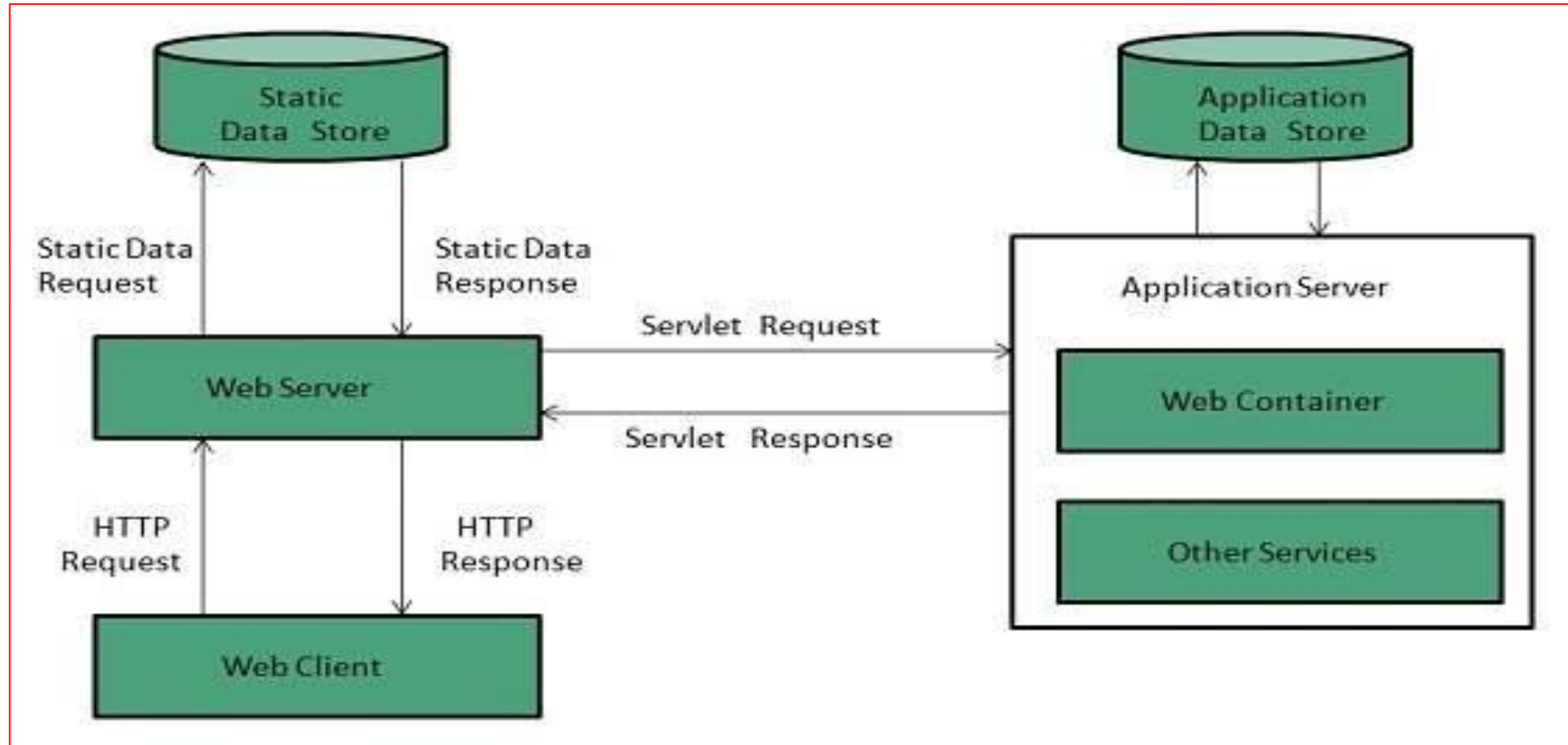
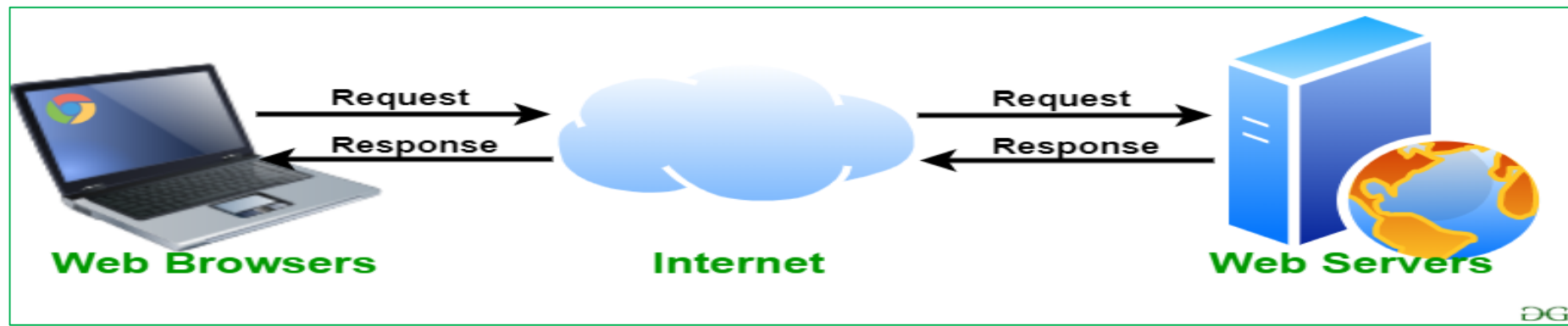
A web client is software that accesses a web server by sending an HTTP request message and processing the resulting HTTP response.

- A web client is an application that communicates with a web server, using Hypertext Transfer Protocol (HTTP). Any web client that is designed to directly support user access to web servers is known as a user agent.
- Browser is the most commonly used web client. The most common interface to the World Wide Web is a browser, such as Mosaic, Netscape Navigator, or Internet Explorer.



➤ Web Server

- Web server is a program which processes the network requests of the users and serves them with files that create web pages. This exchange takes place using Hypertext Transfer Protocol (HTTP).
- Basically, web servers are computers used to store HTTP files which makes a website and when a client requests a certain website, it delivers the requested website to the client. For example, you want to open Facebook on your laptop and enter the URL in the search bar of google. Now, the laptop will send an HTTP request to view the facebook webpage to another computer known as the webserver. This computer (webserver) contains all the files (usually in HTTP format) which make up the website like text, images, gif files, etc. After processing the request, the webserver will send the requested website-related files to your computer and then you can reach the website.
- Different websites can be stored on the same or different web servers but that doesn't affect the actual website that you are seeing in your computer. The web server can be any software or hardware but is usually a software running on a computer. One web server can handle multiple users at any given time which is a necessity otherwise there had to be a web server for each user and considering the current world population, is nearly close to impossible. A web server is never disconnected from the internet because if it was, then it won't be able to receive any requests, and therefore cannot process them.



Key Points

When client sends request for a web page, the web server search for the requested page if requested page is found then it will send it to client with an HTTP response.

If the requested web page is not found, web server will the send an HTTP response:Error 404 Not found.

If client has requested for some other resources then the web server will contact to the application server and data store to construct the HTTP response.

Architecture

Web Server Architecture follows the following two approaches:

1. Concurrent Approach
2. Single-Process-Event-Driven Approach.

Concurrent Approach

Concurrent approach allows the web server to handle multiple client requests at the same time. It can be achieved by following methods:

- Multi-process
- Multi-threaded
- Hybrid method.
- **Multi-processing**
In this a single process (parent process) initiates several single-threaded child processes and distribute incoming requests to these child processes. Each of the child processes are responsible for handling single request. It is the responsibility of parent process to monitor the load and decide if processes should be killed or forked.
- **Multi-threaded**
Unlike Multi-process, it creates multiple single-threaded process.
- **Hybrid**
It is combination of above two approaches. In this approach multiple process are created and each process initiates multiple threads. Each of the threads handles one connection. Using multiple threads in single process results in less load on system resources.

S.N.	Web Server Description
1	Apache HTTP Server This is the most popular web server in the world developed by the Apache Software Foundation. Apache web server is an open source software and can be installed on almost all operating systems including Linux, UNIX, Windows, FreeBSD, Mac OS X and more. About 60% of the web server machines run the Apache Web Server.
2.	Internet Information Services (IIS) The Internet Information Server (IIS) is a high performance Web Server from Microsoft. This web server runs on Windows NT/2000 and 2003 platforms (and may be on upcoming new Windows version also). IIS comes bundled with Windows NT/2000 and 2003; Because IIS is tightly integrated with the operating system so it is relatively easy to administer it.
3.	Lighttpd The lighttpd, pronounced lighty is also a free web server that is distributed with the FreeBSD operating system. This open source web server is fast, secure and consumes much less CPU power. Lighttpd can also run on Windows, Mac OS X, Linux and Solaris operating systems.
4.	Sun Java System Web Server This web server from Sun Microsystems is suited for medium and large web sites. Though the server is free it is not open source. It however, runs on Windows, Linux and UNIX platforms. The Sun Java System web server supports various languages, scripts and technologies required for Web 2.0 such as JSP, Java Servlets, PHP, Perl, Python, and Ruby on Rails, ASP and Coldfusion etc.
5.	Jigsaw Server Jigsaw (W3C's Server) comes from the World Wide Web Consortium. It is open source and free and can run on various platforms like Linux, UNIX, Windows, and Mac OS X Free BSD etc. Jigsaw has been written in Java and can run CGI scripts and PHP

1.2.Client Server Architecture: Single Tier, Two-Tier, Multi-Tier

- The Client-server model is a distributed application structure that partitions task or workload between the providers of a resource or service, called servers, and service requesters called clients.
- In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and deliver the data packets requested back to the client.
- Clients do not share any of their resources. Examples of Client-Server Model are Email, World Wide Web, etc

❑ How the Client-Server Model works ?

1. **Client:** When we talk the word **Client**, it mean to talk of a person or an organization using a particular service. Similarly in the digital world a **Client** is a computer (**Host**) i.e. capable of receiving information or using a particular service from the service providers (**Servers**).

2. **Servers:** Similarly, when we talk the word **Servers**, It mean a person or medium that serves something.

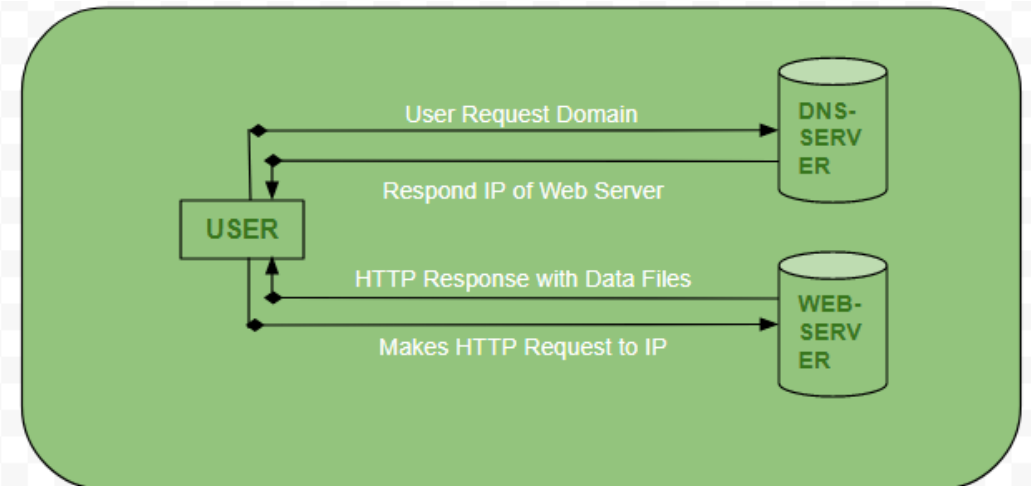
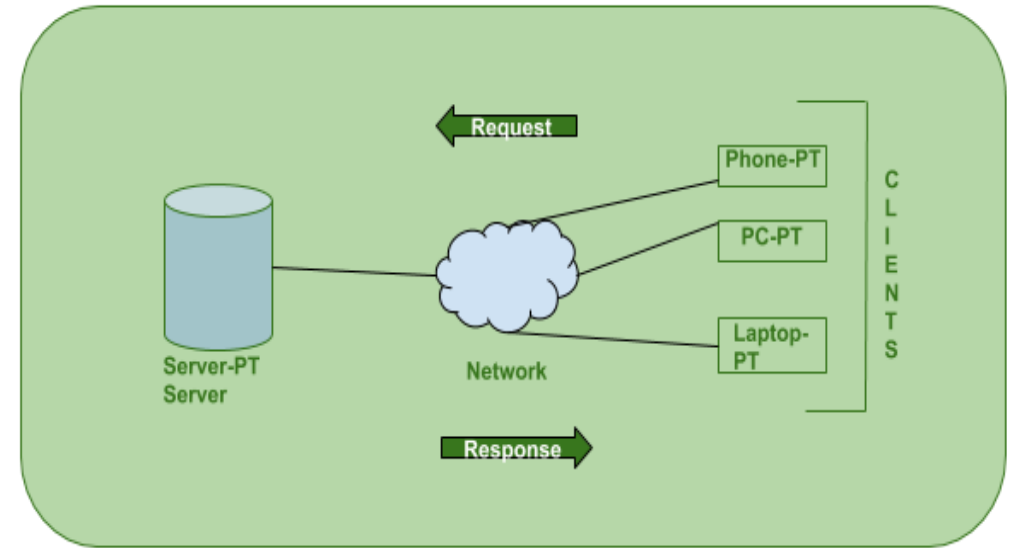
Similarly in this digital world a **Server** is a remote computer which provides information (data) or access to particular services.

So, its basically the **Client** requesting something and the **Server** serving it as long as its present in the database.

❑ How the browser interacts with the servers ?

There are few steps to follow to interact with the servers a client.

- User enters the **URL**(Uniform Resource Locator) of the website or file. The Browser then requests the **DNS**(DOMAIN NAME SYSTEM) Server.
- **DNS Server** lookup for the address of the **WEB Server**.
- **DNS Server** responds with the **IP address** of the **WEB Server**.
- Browser sends over an **HTTP/HTTPS** request to **WEB Server's IP** (provided by **DNS server**).
- Server sends over the necessary files of the website.
- Browser then renders the files and the website is displayed. This rendering is done with the help of **DOM** (Document Object Model) interpreter, **CSS** interpreter and **JS Engine** collectively known as the **JIT** or (Just in Time) Compilers.



❑ **Advantages of Client-Server model:**

- Centralized system with all data in a single place.
- Cost efficient requires less maintenance cost and Data recovery is possible.
- The capacity of the Client and Servers can be changed separately.

❑ **Disadvantages of Client-Server model:**

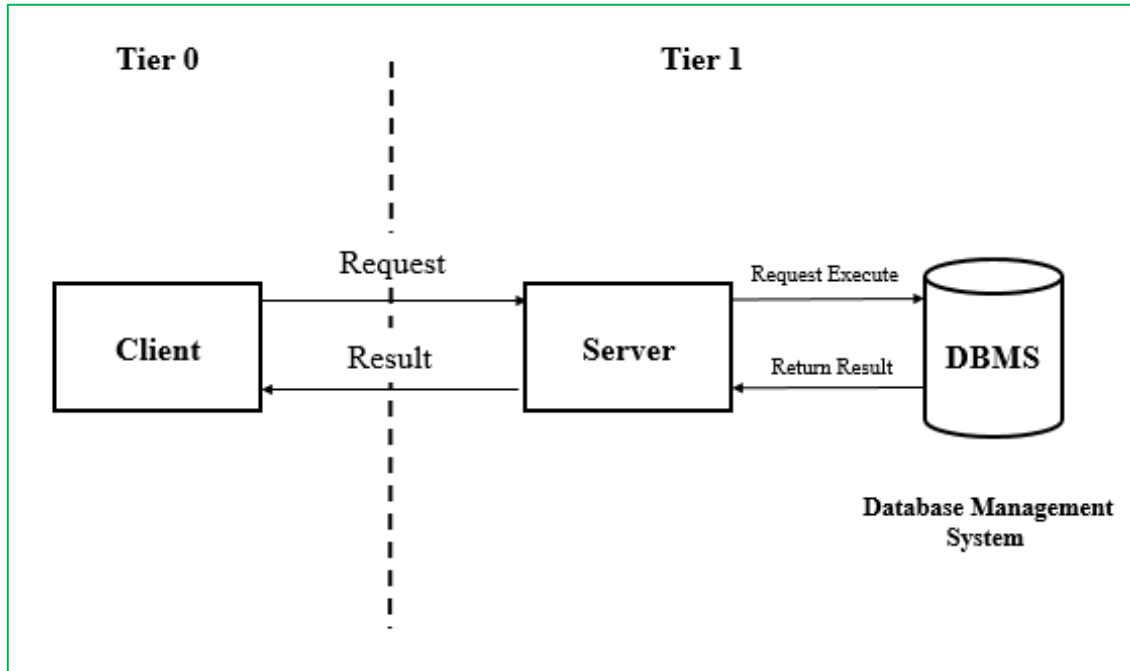
- Clients are prone to viruses, Trojans and worms if present in the Server or uploaded into the Server.
- Server are prone to Denial of Service (DOS) attacks.
- Data packets may be spoofed or modified during transmission.
- Phishing or capturing login credentials or other useful information of the user are common and MITM(Man in the Middle) attacks are common.

❑ Client/Server architecture can be of different model based on the number of layers it holds. Some of them are;

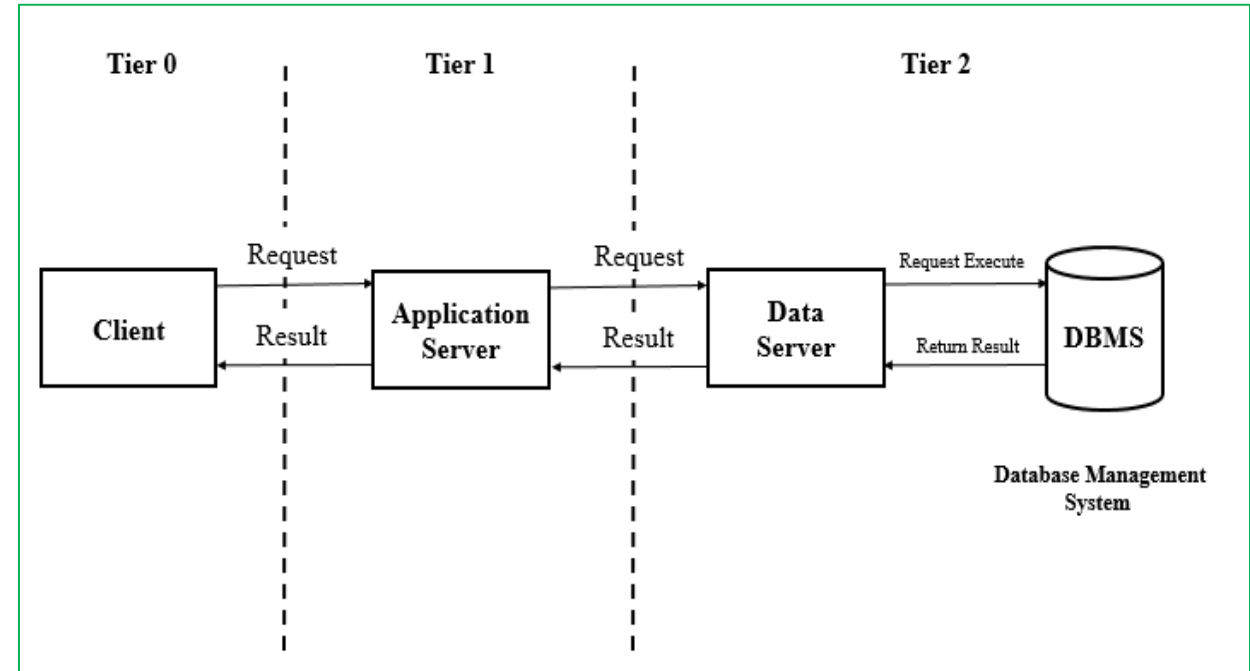
2-Tier Architecture

- 2-tier architecture is used to describe client/server systems where the client requests resources and the server responds directly to the request, using its own resources. This means that the server does not call on another application in order to provide part of the service. It runs the client processes separately from the server processes, usually on a different computer:
- The client processes provide an interface for the customer, and gather and present data usually on the customer's computer. This part of the application is the presentation layer
 - – The server processes provide an interface with the data store of the business. This part of the application is the data layer
 - The business logic that validates data, monitors security and permissions, and performs other business rules can be housed on either the client or the server, or split between the two.
 - Fundamental units of work required to complete the business process
 - Business rules can be automated by an application program.

2-Tier Architecture



3-Tier Architecture



3-Tier Architecture

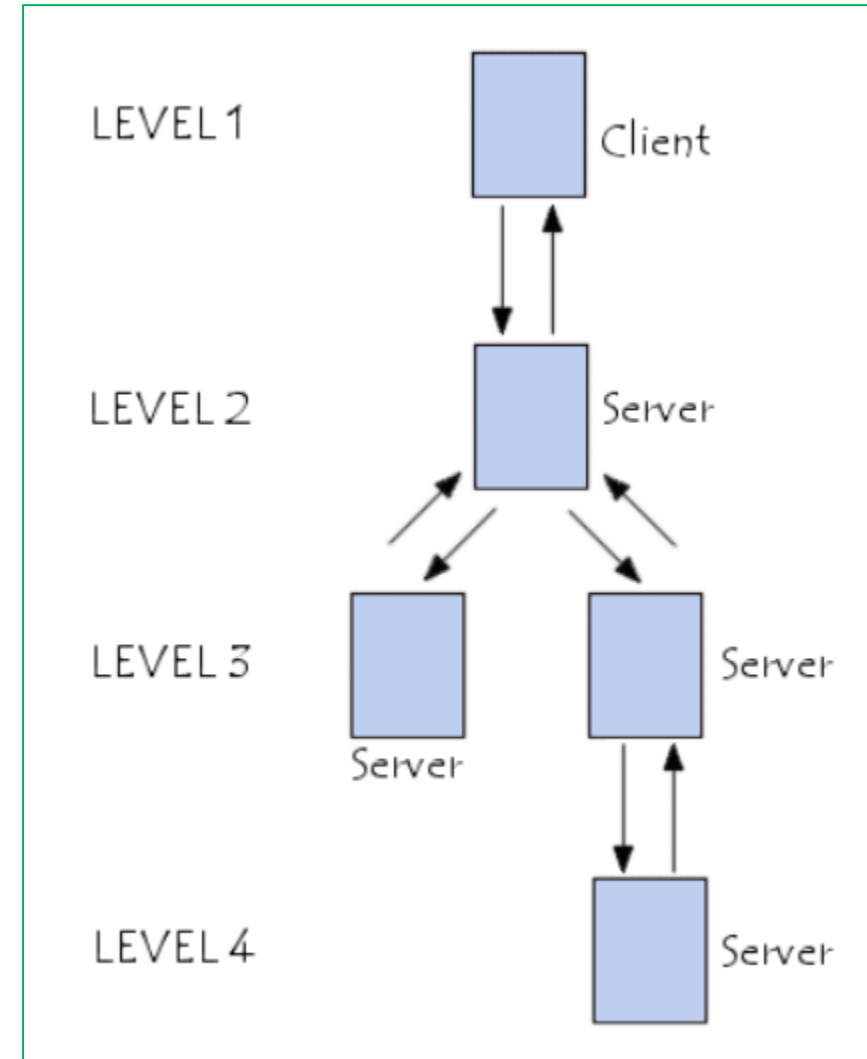
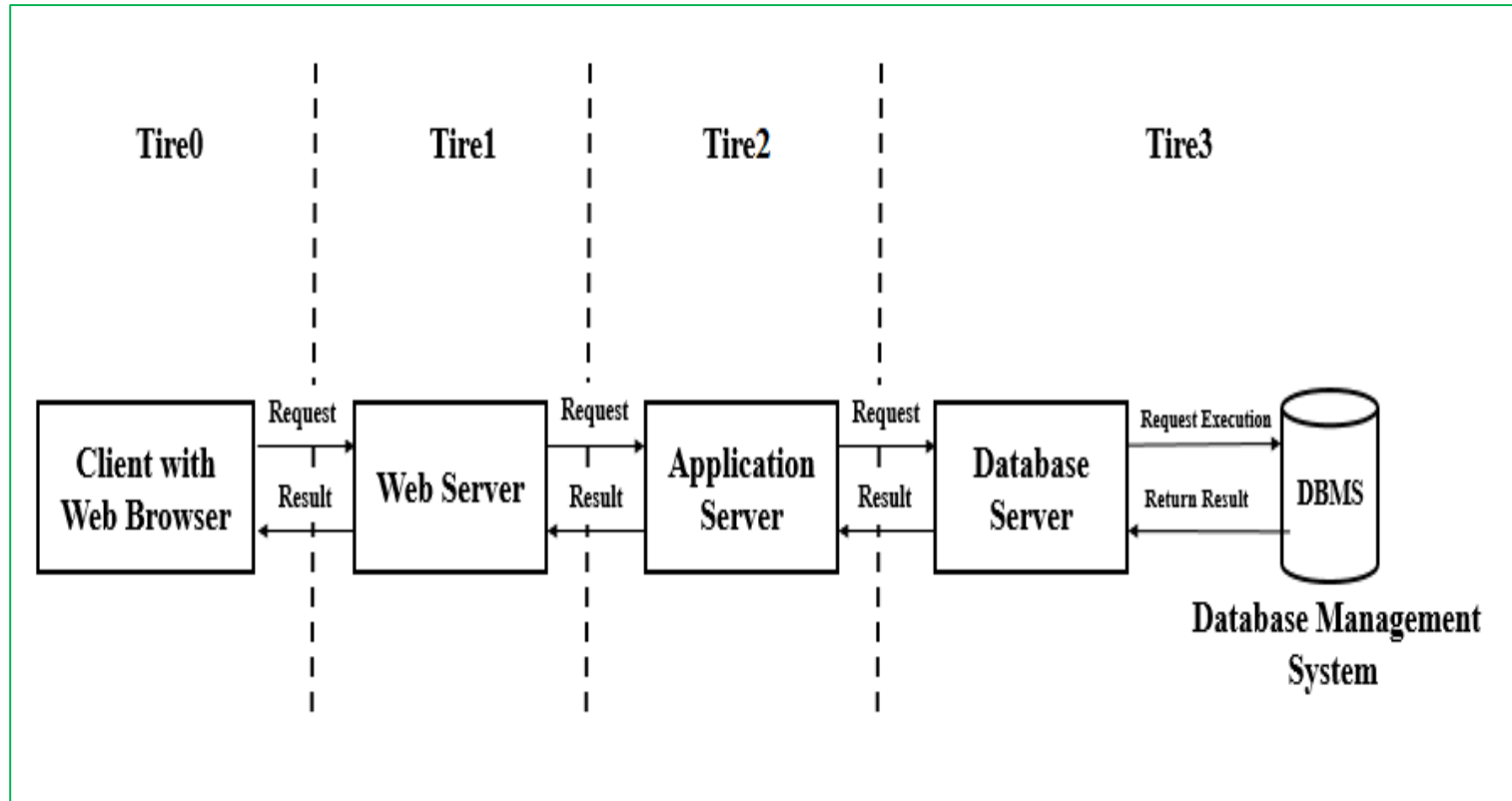
In 3-tier architecture, there is an intermediary level, meaning the architecture is generally split up between:

- A client, i.e. the computer, which requests the resources, equipped with a user interface (usually a web browser) for presentation purposes
- The application server (also called middleware), whose task it is to provide the requested resources, but by calling on another server
- The data server, which provides the application server with the data it requires

N-Tier Architecture (multi-tier)

- N-tier architecture (with N more than 3) is really 3 tier architectures in which the middle tier is split up into new tiers.
- The application tier is broken down into separate parts. What these parts are differs from system to system. The following picture shows it:
- The primary advantage of N-tier architectures is that they make load balancing possible.
- Since the application logic is distributed between several servers, processing can then be more evenly distributed among those servers. N-tiered architectures are also more easily scalable, since only servers experiencing high demand, such as the application server, need be upgraded.
- The primary disadvantage of N-tier architectures is that it is also more difficult to program and test an N-tier architecture due to its increased complexity.

N-Tier Architecture (multi-tier)



Advantages of Multi-Tier Client/Server architectures include:

- Changes to the user interface or to the application logic are largely independent from one another, allowing the application to evolve easily to meet new requirements.
- Network bottlenecks are minimized because the application layer does not transmit extra data to the client, only what is needed to handle a task.
- The client is insulated from database and network operations. The client can access data easily and quickly without having to know where data is or how many servers are on the system.
- Database connections can be 'pooled' and thus shared by several users, which greatly reduces the cost associated with per-user licensing.
- The organization has database independence because the data layer is written using standard SQL which is platform independent. The enterprise is not tied to vendor- specific stored procedures.
- The application layer can be written in standard third or fourth generation languages, such as ASP, PHP with which the organization's in-house programmers are experienced.

What kind of systems can benefit?

- Generally, any Client/Server system can be implemented in an 'N-Tier' architecture, where application logic is partitioned among various servers.
- This application partitioning creates an integrated information infrastructure which enables consistent, secure, and global access to critical data.
- A significant reduction in network traffic, which leads to faster network communications, greater reliability, and greater overall performance is also made possible in a 'N-Tier' Client/Server architecture.

1.3.HTTP: HTTP Request and Response; URL

- The Hypertext Transfer Protocol (HTTP) is an application-level protocol for distributed, collaborative, hypermedia information systems.
- This is the foundation for data communication for the World Wide Web (i.e. internet) since 1990. HTTP is a generic and stateless protocol which can be used for other purposes as well using extensions of its request methods, error codes, and headers.
- This tutorial is based on RFC-2616 specification, which defines the protocol referred to as HTTP/1.1. HTTP/1.1 is a revision of the original HTTP (HTTP/1.0).
- A major difference between HTTP/1.0 and HTTP/1.1 is that HTTP/1.0 uses a new connection for each request/response exchange, where as HTTP/1.1 connection may be used for one or more request/response exchanges.

Basic Features of HTTP

❑ There are three basic features that make HTTP a simple but powerful protocol:

1. HTTP is connectionless:

The HTTP client, i.e., a browser initiates an HTTP request and after a request is made, the client waits for the response. The server processes the request and sends a response back after which client disconnect the connection. So client and server knows about each other during current request and response only. Further requests are made on new connection like client and server are new to each other.

2. HTTP is media independent:

It means, any type of data can be sent by HTTP as long as both the client and the server know how to handle the data content. It is required for the client as well as the server to specify the content type using appropriate MIME-type.

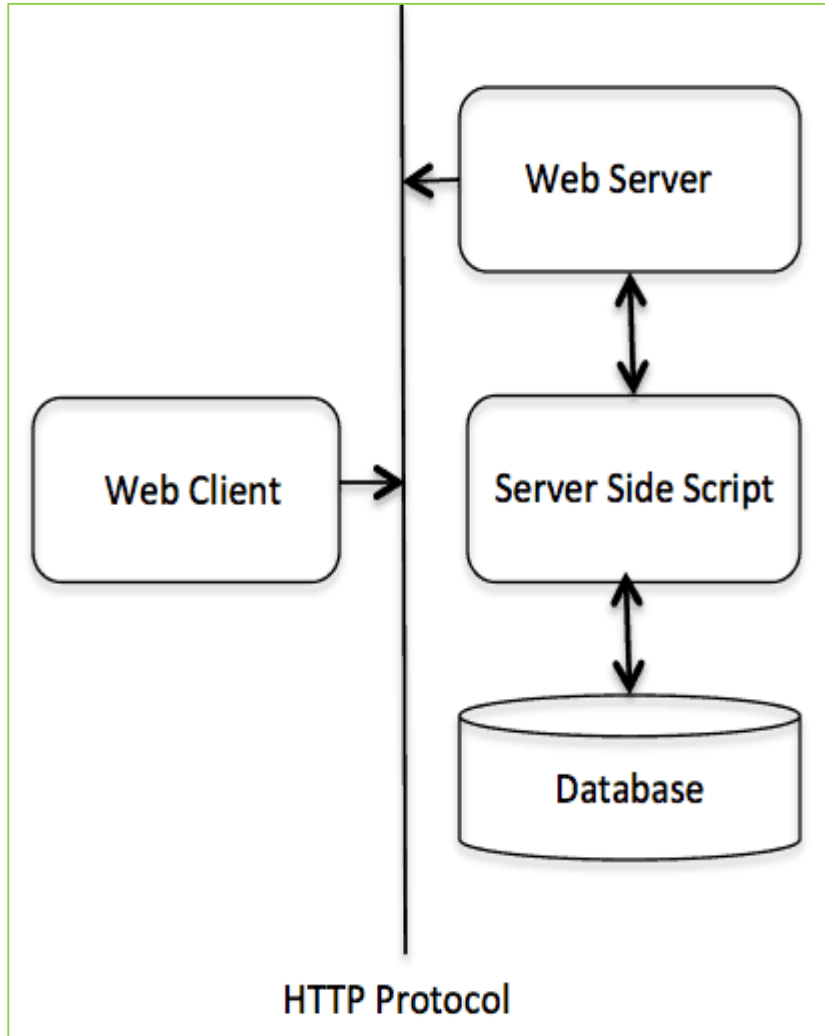
3. HTTP is stateless:

As mentioned above, HTTP is connectionless and it is a direct result of HTTP being a stateless protocol. The server and client are aware of each other only during a current request. Afterwards, both of them forget about each other. Due to this nature of the protocol, neither the client nor the browser can retain information between different requests across the web pages.

HTTP/1.0 uses a new connection for each request/response exchange, where as HTTP/1.1 connection may be used for one or more request/response exchanges.

Basic Architecture Of HTTP

The following diagram shows a very basic architecture of a web application and depicts where HTTP sits:



- The HTTP protocol is a request/response protocol based on the client/server based architecture where web browsers, robots and search engines, etc. act like HTTP clients, and the Web server acts as a server.

Client

The HTTP client sends a request to the server in the form of a request method, URI, and protocol version, followed by a MIME-like message containing request modifiers, client information, and possible body content over a TCP/IP connection.

Server

The HTTP server responds with a status line, including the message's protocol version and a success or error code, followed by a MIME-like message containing server information, entity meta information, and possible entity-body content.

HTTP - Requests

➤ An HTTP client sends an HTTP request to a server in the form of a request message which includes following format:

- A Request-line
- Zero or more header (General | Request | Entity) fields followed by CRLF
- An empty line (i.e., a line with nothing preceding the CRLF) indicating the end of the header fields
- Optionally a message-body

